

Wormhole Foundation EVM-NTT Audit Summary

Prepared by Cyfrin

Version 2.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Wormhole Foundation on Wormhole Foundation EVM-NTT. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated April 11, 2024, was delivered to Wormhole Foundation, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 20 days, the Cyfrin team conducted an audit on the Wormhole Foundation EVM-NTT code provided by Wormhole Foundation. The findings consist of 2 High, 8 Medium & 5 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Wormhole Foundation EVM-NTT
Repository	example-native-token-transfers
Commit	f4e2277b3583...
Audit Timeline	Feb 26th - Mar 22nd, 2024
Methods	Manual Review, Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	2	2	0
Medium Risk	8	5	3
Low Risk	5	3	2
Informational	6	2	4
Gas Optimizations	4	1	3
Total	25	13	12

6 Audit Scope

Cyfrin conducted an audit of the Wormhole Native Token Transfers (NTT) based on the code present in the repository commit hash f4e2277.

The following directories were included in the scope of the audit:

- evm/src/NttManager/*
- evm/src/Transceiver/*
- evm/src/interfaces/*
- evm/src/libraries/*
- evm/src/wormhole/*

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Wormhole is a generic message passing protocol that enables communication between blockchains. Native Token Transfers (NTT) is a framework designed to facilitate the transfer of tokens across different blockchain networks without relying on liquidity pools, offering an open, flexible, and composable solution.

Integrators configure Wormhole NTTs on each chain, including the token standard and metadata. It supports a "locking" mode for existing token deployments to preserve the original token supply on a single chain, and a "burning" mode for deploying natively multichain tokens with a distributed supply.

Key components:

- **Transceivers:** responsible for sending NTT transfers, managed by the NttManager contract on the source chain and delivered to the corresponding NttManager peer on the recipient chain.
- **NttManager:** manages tokens and Transceivers, including rate-limiting and message attestation logic.

Transfers can be rate-limited on both source and destination chains, with excess transfers queued for delayed execution. NTT Managers can use Wormhole Transceivers or implement customized versions.

9 Attestation

Cyfrin attests that it performed a security review of the Wormhole Foundation EVM-NTT code provided by Wormhole Foundation.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-04-11-cyfrin-wormhole-evm-ntt-v2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Wormhole Foundation has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=401be76385134843f3979bebe9d7f585c5b9326d27ba7b535153bcdb49300eb1
F=2024-04-11-cyfrin-wormhole-evm-ntt-v2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.